MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 6942

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal

See figure 1

Mounting position: Vertical

Weight: 3.6 pounds nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ecl	Ec2	Icl	Ib
Unit:	MHz	Vac	Vdc	Vdc	Vdc	mAdc	mAdc
Maximum:	900	6.0	4,000	-150	600	100	600
Minimum:	---	---	---	---	---	---	---
TEST CONDITIONS:	---	5.7 (see note 2)	2,000	Adjust	500	---	200

ABSOLUTE RATINGS:

Parameter:	ik	Pg2	Pp	Pi	tk	T (seal)
Unit:	a	W	W	W	sec	°C
Maximum:	3	15	1,500	2,400	60	See note 1
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	60	See note 3

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
---	Visual and mechanical inspection criteria	-		---	---	---	---	---	---
---	Dimensions	-	See figure 1	1.0	II	---	---	---	---
1266	Total grid current	-	Eb = 4,000 Vdc; Ec2 = 600 Vdc; Ec1/Ib = 375 mAdc; 1st test after holding period	0.65	II	Ic1	---	-20.0	µAdc
1236	Power oscillation	4	Eb = 4,000 Vdc; Ec2 = 600 Vdc (max); Ib = 600 mAdc (max); Rg = 2,000 ohms; t = 5 minutes; F = 850 ±50 MHz; Ic1 = 80 mAdc (max); Ic2 = 25 mAdc (max)	0.65	II	Po	1,000	---	W (useful)
1231	Pulsing emission sinusoid	-	ec1 = ec2 = eb = 750v; tp = 2 to 6 µs; prr = 60	0.65	II	Is	3.5	---	a
1301	Heater current	-		0.65	II	If	22.0	26.0	Aac
1256	Electrode (g2) current	-	Ib = 100 mAdc; Eb = Ec2 = 250 Vdc	0.65	II	Ic2	---	40	mAdc
1261	Electrode (g1) voltage (1)	-		0.65	II	Ec1	-6.0	-20.0	Vdc
1261	Electrode (g1) voltage (2)	-	Ib = 10 mAdc	0.65	II	Ec1	-25.0	-45.0	Vdc
	Quality conformance inspection, part 2								
---	Dimension	-	See figure 1	6.5	S3	---	---	---	---
1266	Primary (g1) grid emission	-	Ef = 6.0 V; Ic1 = 300 mAdc; t = 60 seconds; anode, (g2), and cathode grounded	6.5	S3	Isg1	---	30	µAdc
1266	Primary (g2) grid emission	-	Ef = 6.0 V; Ic2 = 75 mAdc; t = 60 seconds; anode, (g1), and cathode grounded	6.5	S3	Isg2	---	30	µAdc
1306	Transconductance	-	Ib = 250 mAdc	6.5	S3	Gm	14,000	---	µmhos
1316	Amplification factor	6	g2 to g1	6.5	S3	Mu	12.0	22.0	---
1331	Direct-interelectrode capacitance	-	g1 tied to g2; anode grounded	6.5	S3	Cin	15.0	18.5	pF
		-	g1 tied to g2; cathode grounded	6.5	S3	Cout	5.0	6.0	pF
		5	g1 tied to g2	6.5	S3	Cpk	---	0.006	pF
(E) 1031	Low-frequency vibration	8	Ef = 5.7 Vac; Ebb = 275 Vdc; Ec2 = 250 Vdc; Rp = 2,000 ohms; Ec1/Ib = 10 mAdc	---	---	Ep	---	1.0	Vac

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THOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Life test	3, 4, 7, 9	Group D; Ef = 5.5 Vac; power oscillation; t = 500 hours	---	---	---	---	---	---
---	Life - test end points:								
1236	Power oscillation	9		---	---	Po	1,000	---	W (useful)
1231	Pulsing emission sinusoid	9		---	---	is	3.5	---	a
1266	Primary (g1) grid emission	9		---	---	Isg1	---	100	µAdc
1266	Primary (g2) grid emission	9		---	---	Isg2	---	100	µAdc

NOTES:

1. Airflow at sea level, through radiator.

Airflow	60 cfm, minimum.
Static pressure	1.5 inches water.
Heater to cathode seals	8 cfm, minimum.
Screen-grid to control-grid seals	4 cfm, minimum.
Anode to screen-grid ceramic insulator	6 cfm, minimum.
Incoming air temperature	45°C, maximum.
Radiator hub temperature at fin adjacent to anode seal	180°C, maximum.
Tube temperature at any point	200°C, maximum.

   a. The volume of cooling air indicated for the various seals is approximate only. Distribution of cooling air will vary with the cavity configuration about the tube. For satisfactory operation, the maximum temperature of any point on the tube shall be below 200°C.
   b. Forced-air cooling shall be applied simultaneously with the application of any voltages and shall be maintained for 1 minute after the removal of all voltages.

2. Heater voltage shall be applied so that starting heater current does not exceed 36 amperes, even momentarily.

3. Cooling air shall be provided to limit ceramic temperatures at any point to 200°C maximum, and also to limit the radiator hub temperature at fin adjacent to anode seal to 180°C maximum. For satisfactory operation, the airflow need not be greater than 125 percent of the minimum values as specified in note 1. Conditions of note 1, item (b) shall apply.

4. Cavity as shown on Drawing 286-JAN, or equivalent, shall be used.

5. Measurement shall be made with complete external shielding between cathode and anode as shown on Drawing 287-JAN.

6. Tube shall be operated with Ef = 5.7 V, Eb = 2,000 Vdc, Ib = 200 mAdc. Record difference of Ec1 with Ec2 set at 500 to 450 Vdc. Amplification factor is equal to the difference in Ec1 values divided into 50.

7. The tube on life test shall be cycled seven times per week by turning the equipment "off" for a minimum of half an hour to cool, then turning it "on" again. Minimum "on" time shall be 3 hours with a maximum of two cycles per day.

8. Tube shall be mounted by socket head cap screw holes (handle) in a vertical position.

(E) 9. Service-life guarantee is acceptable in lieu of life test if specifically authorized by contract.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 85
DLA - ES

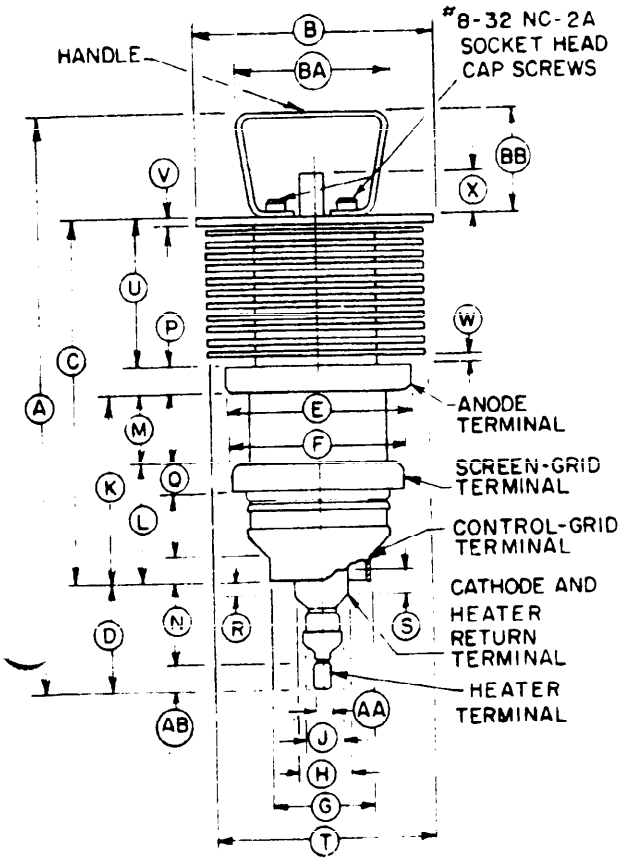
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	8.250 (209.35)	8.750 (222.25)
C	5.375 (136.53)	5.625 (142.88)
D	1.406 (35.71)	1.531 (38.89)
K	2.813 (71.45)	2.938 (74.63)
L	1.688 (42.88)	1.813 (46.05)
M	1.109 (28.17)	1.141 (28.98)
Q	.344 (8.74)	.422 (10.72)
R	.156 (3.96)	.281 (7.14)
S	.375 (9.53)	.438 (11.13)
W	.090 (2.29)	.130 (3.30)
X		.750 (19.05)
Quality conformance inspection, part 2		
B	3.484 (88.49)	3.516 (89.31)
E	2.745 (69.72)	2.765 (70.23)
F	2.515 (63.88)	2.535 (64.39)
G	1.522 (38.66)	1.542 (39.17)
H	.652 (16.58)	.668 (16.97)
J		.438 (11.13)
N	.359 (9.12)	.391 (9.93)
P	.359 (9.12)	.391 (9.93)
T	3.240 (82.30)	3.260 (82.80)
U	2.156 (54.76)	2.406 (61.11)
V	.090 (2.29)	.130 (3.30)
AA	.242 (6.15)	.258 (6.55)
AB	.365 (9.27)	.385 (9.78)
BA		2.375 (60.33)
BB		1.625 (41.28)

NOTES:
a. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
b. Permissible eccentricity of diameters with respect to centerline through anode terminal and control-grid terminal shall be as follows:

Diameter	Eccentricity, max
T	.010
H	.015
AA	.030

FIGURE 1. Outline drawing of electron tube type 6942.

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